

PAPER_1087_ERRATUM

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PAPER_1087 ERRATUM — Unit Inconsistency in Abstract Formula vs §3 Table

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Subject: Internal inconsistency between the abstract closed form and the §3 numerical table in PAPER_1087 (Time-Evolving Dark Energy EOS w_{DE} from SCm Phonon Dynamics).

Status: OPEN — awaiting clarification of units; calculator closure pinned to §3 table value at present ($t = 13.8$ Gyr).

The Discrepancy

PAPER_1087 abstract gives:

``

$$w_{DE}(t, \Gamma) = -1 + (2\kappa t + (SSq/26) \cdot t) / \ln(\Phi(\Gamma))$$

``

with $\kappa = 5.787 \times 10^{-9} \text{ s}^{-1}$, $SSq = 0.57$, $\Phi(\Gamma) = \Phi_{26} \cdot S_{26}$, $\Phi_{26} = 10^2$.

Direct evaluation at $t = 4.35 \times 10^{10} \text{ s}$ (≈ 13.8 Gyr) with κ in s^{-1} :

``

$$\text{Numerator} \approx 2 \times 5.787\text{e-}9 \times 4.35\text{e}17 + (0.57/26) \times 4.35\text{e}17 \approx 9.54 \times 10^{10}$$

$$\ln(\Phi) = \ln(1.453 \times 10^2) = 46.43$$

$$w_{DE} = -1 + 9.54\text{e}10 / 46.43 \approx 2.05 \times 10^1$$

``

This is non-physical (w_{DE} should be near -1).

Conversely, the **§3 table** in the paper reports:

| t (Gyr) | w_{DE} | Δw |

|---|---|---|

| 0 | -1.0000 | 0 |

| 1 | -0.9959 | 0.0041 |

| 5 | -0.9795 | 0.0205 |

| 10 | -0.9591 | 0.0409 |

| **13.8** | **-0.9435** | **0.0565** |

For the table values to be reproduced, the deviation at $t = 13.8$ Gyr must be $\Delta w = 0.0565$. Solving backward:

``

$$\text{required coefficient } C \text{ such that } C \times 13.8 / \ln(\Phi) = 0.0565$$

$$C = 0.0565 \times 46.43 / 13.8 = 0.190 \text{ per Gyr}$$

``

Neither the κ term ($2 \times 5.787\text{e-}9 = 1.16\text{e-}8$ per Gyr) nor the SSq/26 term (0.0219) reaches 0.190 — short by a factor of ~ 9 .

Three candidate resolutions

1. **κ in different units.** If κ is meant to be $5.787 \times 10^{\blacksquare} \times 10^{\blacksquare} = 0.058$ per Gyr (perhaps the $10^{\blacksquare}$ is a typo for $10^{\blacksquare^2}$ in some natural unit system), then $2\kappa \approx 0.116$ — closer but still short.
2. **(SSq/26) should be (SSq $\times$ X)** where $X \approx 8.5$. The closest UQFF candidates that give ~ 8.5 are $K_{\text{MEX}} \times A_5 / SO_5 / 1.47$ or similar, but none is structurally exact.
3. **Formula should be applied with different normalization.** Perhaps the t in the formula is "dimensionless cosmic time" and the table substitutes a different scaling.

Calculator behavior (current)

`calculate_paradox({"paradox": "dark_energy_eos_time_evolvering"})` and `calculate_cosmology({"observable": "w_de_paper_1087"})` both return **-0.9435** (the §3 table value at $t = 13.8$ Gyr) directly. The literal formula is still documented in the closure's other fields for transparency.

Recommended action by Daniel

- Confirm κ units and provide an erratum to the paper itself; OR
- Replace (SSq/26) with the correct coefficient; OR
- Confirm that the closure should remain pinned to the §3 table value pending re-derivation.

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